

Bebop Router Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Bebop on Router. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated June 12, 2026, was delivered to Bebop, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Router code provided by Bebop. The findings consist of 1 High, 2 Medium & 18 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Router
Repository	bebop-rfqa
Commit	528826acf28d...
Fix Commit	9db95bc2b423...
Audit Timeline	June 2nd - June 8th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	1	0
Medium Risk	2	2	0
Low Risk	18	10	8
Informational	4	1	3
Gas Optimizations	1	1	0
Total	26	15	11

6 Audit Scope

The audit scope was limited to:

```
contracts/BebopRouter.sol
contracts/Errors.sol
contracts/libraries/BebopRouterOrderLib.sol
contracts/libraries/HookLib.sol
contracts/base/BebopPmmHelper.sol
contracts/base/BebopValidation.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

`BebopRouter` is a request-for-quote (RFQ) swap router that executes maker-signed quotes by wrapping the external Bebop private market maker (`BebopSettlement`): it pulls the taker's input, adjusts the fill amount and forwards the settlement call, then distributes the output between the receiver, the order's makers and the protocol treasury.

On top of the base settlement it adds:

- per-order fee and slippage pricing through pluggable `IChecker` and `IOracle` contracts named in each order
- pre and post execution hooks
- native-token wrapping and unwrapping
- a gasless `settle` path that pulls user funds via a Permit2 witness signature or an EIP-712 / ERC-1271 order signature

The router is deployed on Ethereum mainnet and Polygon.

9 Attestation

Cyfrin attests that it performed a security review of the Router code provided by Bebop.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-06-12-cyfrin-bebop-router-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Bebop has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=7814f843e7c6df6e244b0ff27c9b758fe6649a58ebd90d00e7b0ff04ec68d4c3
F=2026-06-12-cyfrin-bebop-router-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.